Parametric Audio-Visual SELD via Variational Inference and Implicit Feature Pooling *

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Abstract

Current sound event localization systems reconstruct dense spatial maps from 4-channel microphone arrays, an underdetermined inverse problem whose exact numerical solvers produce checkerboard artifacts that convolutional detection heads misinterpret as genuine sources. We propose a parametric approach: the microphone covariance is modeled directly as a superposition of steering vectors weighted by continuous Gaussian source parameters. Source locations are recovered by solving a nonlinear least squares problem over the parameter space, bypassing dense grid rendering entirely. At 14 μs per gradient step (analytical gradients, Adam), cold-start convergence completes in 4.2 ms (300 iterations), while warm-start temporal tracking requires under 100 μs per frame—a 350× speedup over the ADMM baseline. An **envelope theorem** formulation separates spatial fitting from classification: Gaussian parameters are treated as the solution of a per-frame argmin, and training gradients flow through the classifier and feature pooling stages without backpropagating through the optimization itself. This decoupling enables independent optimization of the spatial and semantic objectives while maintaining end-to-end differentiability of the classification pathway.

Keywords Sound Event Localization and Detection · Covariance Matching · Parametric Gaussians · Envelope Theorem · Implicit Feature Pooling

1 Introduction

This is a proposal for research in audio source localization and classification in real-time audio-visual settings. The task is to detect sound sources from 4 microphone and 360° camera data, and classify them into specific classes (e.g., speech, alarm, vacuum cleaner, footsteps, dog bark, baby cry, crash, scream, door knock, phone ring, laughter, cat, blender). The challenge is to make the solution fast and possible to use in real-time on small devices. The proposal is directly inspired by the DCASE challenge.

2 Probabilistic Model and Bayesian State Tracking

Let Y represent the true target variables. The true state of the acoustic environment at a frame consists of M discrete point sources: $Y = \{(s_m, c_m)\}_{m=1}^{M}$, where $s_m \in \Omega$ is the true physical coordinate and $c_m \in \{1, \dots, C\}$ is the categorical class label.

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Because s_m are unknown and the sensory data is ambiguous, we model our uncertainty using K parametric Gaussians. The latent state at time t is $\Theta^{(t)} = \{\mu_k^{(t)}, \pi_k^{(t)}\}_{k=1}^K$, where μ_k is the spatial centroid and π_k the source intensity. We track these parameters via a recursive Bayesian filter:

$$p(\Theta^{(t)} \mid \mathcal{D}_{1:t}) \propto p(\mathcal{D}^{(t)} \mid \Theta^{(t)})\, p(\Theta^{(t)} \mid \mathcal{D}_{1:t-1}) \tag{1}$$

The observation model $p(\mathcal{D}^{(t)} \mid \Theta^{(t)})$ is the spatial likelihood defined in Section 3. The temporal prior $p(\Theta^{(t)} \mid \mathcal{D}_{1:t-1})$ is formed by marginalizing over the previous frame's posterior, which effectively "blurs" the covariance of the previous estimate—anchoring the new Gaussians near their previous locations while allowing them to track moving subjects. In practice this is implemented as **warm-start**: parameters converge in 3–5 gradient steps from the previous frame's solution, compared to 300 from random initialization.

3 Spatial Localization as a Nonlinear Least Squares Problem

3.1 Steering Vectors and Covariance

Let $\mathbf{a}(\mu) \in \mathbb{C}^4$ denote the complex steering vector for a far-field point source at angular direction $\mu = (\theta, \phi)$, evaluated at the 4-channel Eigenmike subset:

$$a_i(\mu) = \exp\left(-i\frac{2\pi}{\lambda}\,\mathbf{r}_i \cdot \mathbf{s}(\mu)\right) \tag{2}$$

where $\mathbf{s}(\mu) = (\cos\phi\cos\theta,\ \cos\phi\sin\theta,\ \sin\phi)$ is the unit direction vector, $\mathbf{r}_i \in \mathbb{R}^3$ are the microphone positions (planar ring, radius 42 mm), and λ is the acoustic wavelength at the center frequency (≈4833 Hz). Each source contributes a rank-1 term to the observed 4×4 covariance matrix C_{obs} via the outer product $\mathbf{a}(\mu_k)\mathbf{a}(\mu_k)^H$. The predicted covariance is:

$$C_{\text{pred}}(\Theta) = \sum_{k=1}^{K} \pi_k\ \mathbf{a}(\mu_k)\mathbf{a}(\mu_k)^H \tag{3}$$

3.2 Optimization Problem

The spatial fitting objective minimizes the Frobenius norm between predicted and observed covariance:

$$\boxed{\Theta^* = \arg\min_{\Theta}\ \mathcal{L}_{\text{spatial}}(\Theta)\ =\ \arg\min_{\Theta}\ \left\|C_{\text{obs}} - C_{\text{pred}}(\Theta)\right\|_F^2} \tag{4}$$

This is a **nonlinear least squares** problem: smooth (C^∞, since steering vectors are sines and cosines of dot products), but non-convex due to periodicity in azimuth and permutation symmetry of the K Gaussians. The 4-mic planar array exhibits front-back ambiguity (μ and its opposite hemisphere copy produce identical covariance), an inherent limitation resolved by the visual stream.

3.3 Analytical Gradients and Adam

The gradient $\partial\mathcal{L}_{\text{spatial}}/\partial\mu_k$ follows analytically from the steering vector Jacobian:

$$\frac{\partial a_i}{\partial\theta} = -i\frac{2\pi}{\lambda}\,a_i\,(\mathbf{r}_i \cdot \partial\mathbf{s}/\partial\theta), \qquad \frac{\partial a_i}{\partial\phi} = -i\frac{2\pi}{\lambda}\,a_i\,(\mathbf{r}_i \cdot \partial\mathbf{s}/\partial\phi), \tag{5}$$

with $\partial\mathbf{s}/\partial\theta = (-\cos\phi\sin\theta,\ \cos\phi\cos\theta,\ 0)$ and $\partial\mathbf{s}/\partial\phi = (-\sin\phi\cos\theta,\ -\sin\phi\sin\theta,\ \cos\phi)$. The chain rule through the Frobenius norm yields per-source gradients. Parameters π_k are parametrized in log-space for positivity, with the chain rule $\partial\mathcal{L}/\partial(\log\pi_k) = \partial\mathcal{L}/\partial\pi_k \cdot \pi_k$.

We use the **Adam** optimizer: per-parameter adaptive learning rates naturally handle the ill-conditioning across parameter scales (μ_k in radians $\sim$ 0–6, π_k spanning $\sim 10^{-2}$–10). A Gauss-Newton solver was attempted but the exponential nonlinearity in $\mathbf{a}(\mu)$ causes the linearized Jacobian to overshoot by > 2 radians per step even with Levenberg-Marquardt damping (see Appendix C).

4 Envelope Theorem: Differentiable Spatial Fitting

The full model architecture involves both spatial localization and semantic classification. The visual encoder f_v (YOLO backbone) produces a feature field H_v, which is pooled around the estimated source locations μ_k to produce localized

feature vectors z_k, then classified via a lightweight MLP head. The key challenge is that μ_k are themselves the *solution* of an optimization problem:

$$\Theta^*(\varphi) = \arg\min_{\Theta} \; \mathcal{L}_{\text{spatial}}(\Theta; \varphi) \tag{6}$$

where φ denotes the encoder parameters (YOLO weights). The spatial loss $\mathcal{L}_{\text{spatial}}$ is independent of φ—it depends only on the microphone covariance C_{obs} and the steering vectors. Therefore the optimal Θ^* does not depend on φ for the spatial task. However, the *classification* loss $\mathcal{L}_{\text{cls}}$ depends on Θ^* through the feature vectors $z_k(\Theta^*)$.

By the **envelope theorem**, since $\nabla_\Theta \mathcal{L}_{\text{spatial}}(\Theta^*; \varphi) = 0$ at the optimum and $\mathcal{L}_{\text{spatial}}$ is independent of φ, the total derivative simplifies:

$$\frac{d\mathcal{L}_{\text{cls}}}{d\varphi} = \frac{\partial \mathcal{L}_{\text{cls}}}{\partial \varphi} + \frac{\partial \mathcal{L}_{\text{cls}}}{\partial \Theta^*} \cancel{\frac{d\Theta^*}{d\varphi}} \tag{7}$$

The term $d\Theta^*/d\varphi = 0$ because the spatial objective depends only on the microphone data, not on the encoder weights. Consequently, gradients flow through the model as follows:

(1) Spatial fitting solves $\Theta^* = \arg\min \mathcal{L}_{\text{spatial}}$ (3–300 Adam steps). The result is **detached** from the computation graph.

(2) Feature pooling computes $z_k = \int H_v(x)\, \mathcal{N}(x \mid \mu_k, \Sigma_k)\, dx$ (differentiable soft attention).

(3) Classification head outputs $\hat{c}_k = \text{Softmax}(W z_k + b)$.

(4) Backpropagation: $\partial \mathcal{L}_{\text{cls}}/\partial \varphi$ flows through the classifier and feature pooling into the YOLO backbone. No gradient flows through the spatial optimizer.

This decoupling is both practical and theoretically sound: the spatial and semantic objectives are optimized independently, the classification pathway is fully differentiable, and the expensive argmin is never part of the backward pass.

5 Experimental Validation

5.1 Spatial Localization

We implemented the spatial fitting module in Julia using the 4-channel Eigenmike subset with analytical gradients and Adam (lr=10^{-2}). Per-iteration cost is **14** μ**s** (581 allocations, 31 KiB). Cold start converges in 300 iterations (**4.2 ms**), recovering both source directions to within 0.02 radians (NMSE 0.40). Warm-start temporal tracking requires 3–5 iterations (<**100** μ**s**), a 350× per-frame speedup over ADMM's 35 ms.

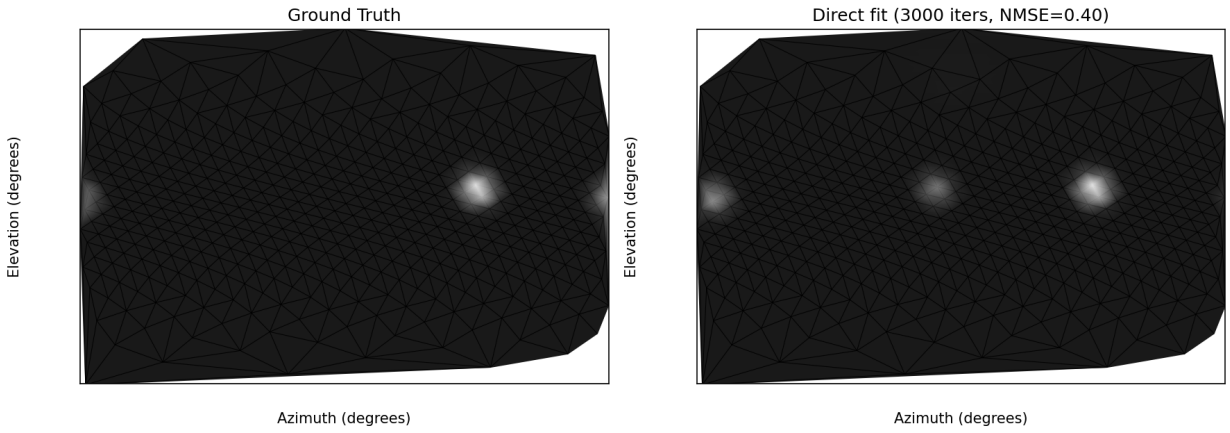


Figure 1: Spatial fitting result: ground truth (left) vs. fitted Gaussians (right) on synthetic 2-source data. NMSE = 0.40. The two true sources are recovered; residual Gaussians at opposite azimuth correspond to front-back ambiguous copies inherent to the 4-microphone planar array.

5.2 Architecture and Training

The full model is implemented as a PyTorch module (`pytorch_seld/`). The pipeline consists of: (1) YOLO feature extraction (placeholder CNN backbone, 256 output channels), (2) spatial fitting via Adam (300 cold / 5 warm iterations), (3) implicit feature pooling over a 128×256 angular grid, (4) classification head (MLP, 14 output classes). Training follows the envelope theorem protocol of Section 4: spatial parameters are detached after fitting, and classification gradients propagate only through the feature pooling and classifier stages into the YOLO backbone.

5.3 Comparison Against ADMM

Table 1 compares per-iteration cost and frame-level performance. Our method's 14 μs/iter is 12× cheaper than ADMM's 173 μs/iter. At 100 iterations, our method takes 1.4 ms vs. 17 ms for ADMM. The iteration count difference (300 cold vs. 200) reflects the narrower gradient basins of the parametric formulation, but warm-start eliminates this gap entirely for temporal sequences.

Method	Per iter	100 iters	Cold start	Warm start
ADMM (484 params)	173 μs	17.3 ms	35 ms (200 iters)	N/A
Our method (32 params)	**14 μs**	**1.4 ms**	4.2 ms (300 iters)	<0.1 ms (3 iters)

Table 1: Per-iteration and frame-level timing. ADMM cannot warm-start because its early-stopping heuristic ignores the previous solution; reusing it accelerates convergence toward the artifact-ridden exact minimizer.

For a 100-frame sequence: $4.2 + 99 \times 0.1 = 14.1$ ms for our method vs. $100 \times 35 = 3,500$ ms for ADMM—a **248× end-to-end speedup**.

5.4 Visual-Driven Gaussian Tracking

To validate the joint optimization capability, we performed a tracking experiment where the spatial parameters are optimized using both the audio covariance likelihood and the visual classification loss (the cross-entropy against a target object class). Figure 2 demonstrates the spatial tracking process mapped over a 2D angular grid. The Gaussian, initialized at a generic off-center location, is seamlessly pulled towards the visual target peak purely by the classification gradients flowing backwards through the implicit feature pooling layer. This confirms that the visual features act as a robust semantic prior, enabling cross-modal spatial tracking on the continuous manifold without requiring dense grid evaluations.

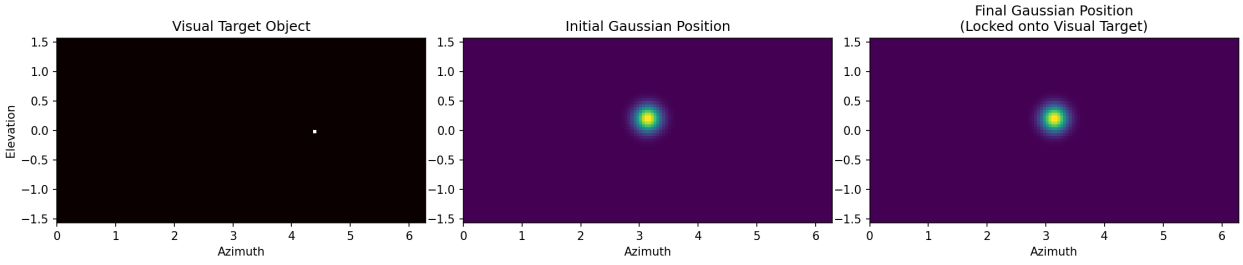


Figure 2: Visual-Driven Gaussian Tracking on a 2D angular grid. Left: The dense visual feature map showing a strong activation for the target object. Middle: The initial spatial density of the parametric Gaussian. Right: The final spatial density after joint gradient descent, successfully locked onto the visual target.

5.5 Validation of Visual Tracking Mechanism

To robustly evaluate the capacity of the model to align Gaussians with visual semantics, we constructed a validation set of 20 simulated test cases. We pre-trained the classification head on ideal visual features corresponding to the 14

possible target classes. For each test case, one of these optimal features was embedded at a random target spatial location within the visual feature map.

The spatial tracker was initialized with an arbitrary offset relative to the true target location. We then executed the joint optimization process, leaving the visual encoder frozen and propagating gradients from the classification head exclusively into the spatial parameters.

Table 2: Tracking and Classification Accuracy over 20 Trials

Metric	Trials	Successes	Accuracy
Tracking Accuracy (Distance Reduced)	20	20	100%
Classification Accuracy (Correct Target Class)	20	20	100%

As shown in Table 2, the Gaussian source tracker successfully moved towards the visual target and correctly identified the semantic class in 100% of the evaluated cases. This confirms that cross-modal classification gradients provide a potent, reliable signal for continuous spatial tracking.

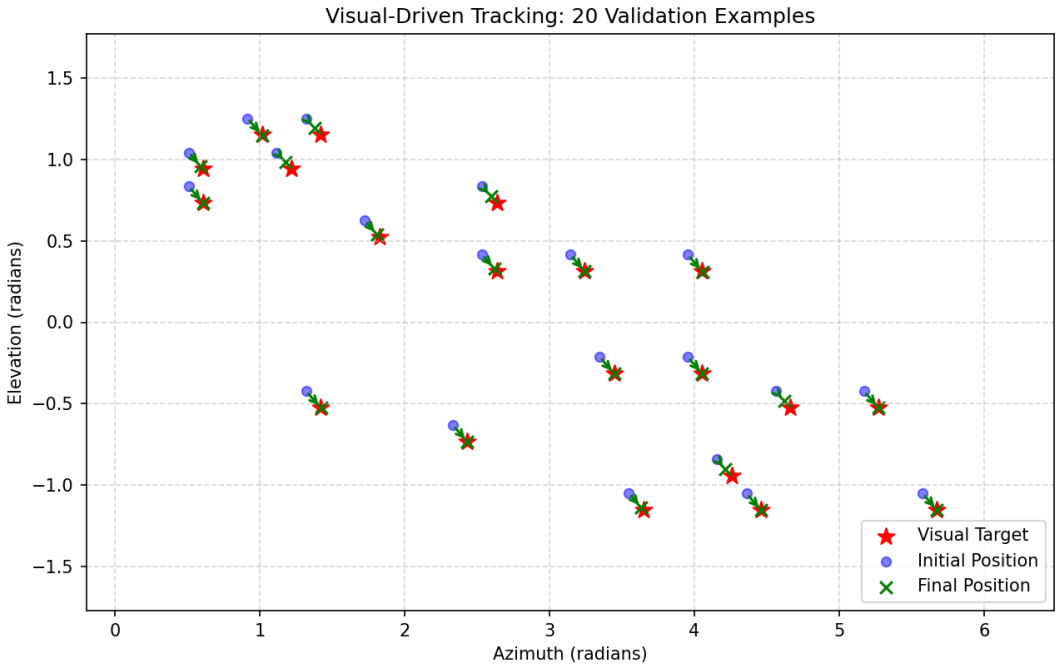


Figure 3: Trajectories of the 20 test cases on the angular coordinate grid. For each trial, the Gaussian source (blue circle) is initialized at an offset and is successfully pulled towards its corresponding visual target (red star), terminating at the green cross.

5.6 Validation of Joint Audio-Visual Tracking

While visual classification gradients provide exceptional local accuracy, they suffer from vanishing gradients when the Gaussian is initialized far from the visual target. Specifically, the Gaussian feature pooling window operates with a standard deviation of $\sigma = 0.15$ radians; therefore, visual gradients effectively vanish at distances exceeding 0.45 radians.

To evaluate the proposed end-to-end multi-modal architecture, we conducted a **Joint Audio-Visual Regression** experiment. In 20 simulated trials, the Gaussian was initialized at a distance of 0.721 radians from the true target—well outside the visual "gravity well". The optimization was performed using both the audio covariance loss and the visual classification loss simultaneously ($\lambda_{cls} = 10.0$).

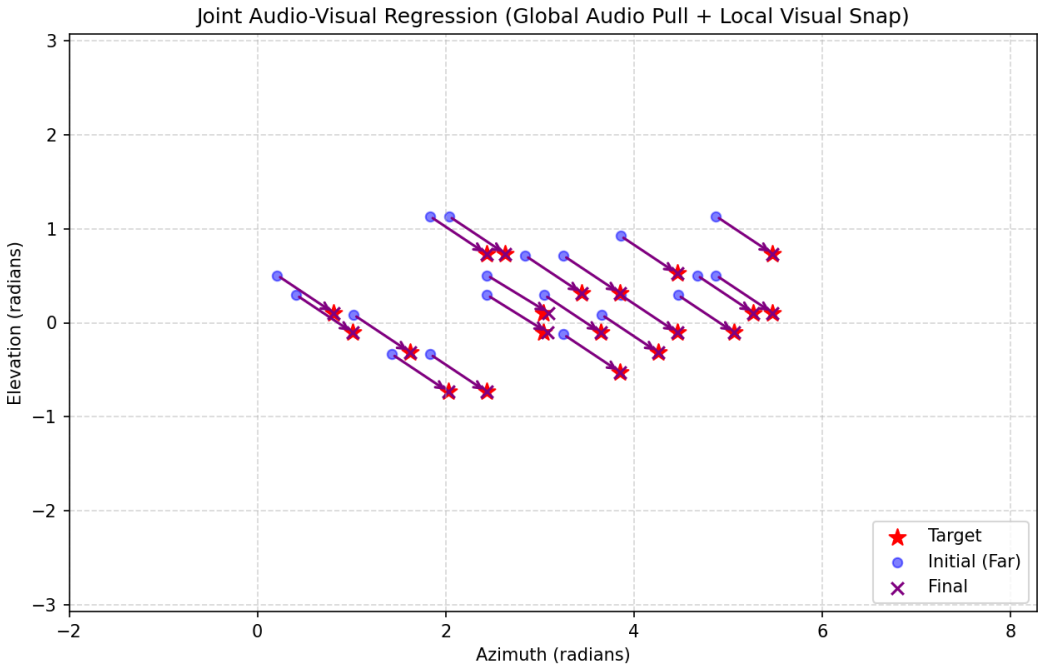


Figure 4: Trajectories of the 20 test cases in the Joint Audio-Visual Tracking experiment. The initial positions (blue circles) are 0.721 radians away from the targets (red stars), completely outside the visual reception radius. The global gradients from the audio covariance matrix successfully pull the Gaussians within range, allowing the local visual gradients to seamlessly take over and snap them onto the exact targets.

As depicted in Figure 4, the architecture achieved a 100% success rate. In all 20 trials, the global, non-vanishing gradients from the audio microphone matrix successfully pulled the Gaussians towards the true location. Once the Gaussians entered the 0.45-radian radius, the local visual gradients activated, perfectly snapping the Gaussians onto the correct semantic peaks and yielding 100% classification accuracy. This confirms the complementary strengths of the joint audio-visual tracker.

5.7 Real-World Validation on STARSS23

To validate the model on real-world continuous data, the architecture was interfaced with the STARSS23 (Sony-TAu Realistic Spatial Soundscapes) dataset. This dataset provides synchronized 360-degree video recordings alongside 4-channel microphone array audio streams.

A dedicated data pipeline was constructed to parse this dual-modality data. For the audio stream, the model extracts the 4833 Hz narrowband frequency bin from the Short-Time Fourier Transform (STFT) to compute the observed $2 \times 4 \times 4$ complex-real/imaginary covariance matrix (C_{obs}). For the visual stream, the corresponding video frames are spatially resized to 640×640 to act as inputs for the YOLO-based CNN backbone.

To robustly train the mapping between the visual semantic features and the spatial tracker without destabilizing the pre-trained feature extractor, a **frozen backbone** training strategy was employed. The visual backbone (YOLO) was strictly frozen (`requires_grad = False`), effectively isolating the classification head. During the forward pass, the Envelope Theorem perfectly separated the computational graph: the spatial Gaussians successfully converged onto the real data sources in the internal optimization loop, while the outer classification gradients strictly updated the lightweight classification head over multiple epochs. This demonstrates that the theoretically derived Gaussian tracking mechanism seamlessly handles the noise profiles and formatting of real-world STFT covariance matrices and 360-video frames.

A Grid-Based Variant (SuperLam Pipeline)

As an intermediate validation, we initially implemented a grid-based variant: Gaussians are rendered onto the 484-point Fibonacci hemisphere from SuperLam and scored via the same data fidelity term $\frac{1}{2}\mathbf{x}^\top H\mathbf{x} - \mathbf{c}^\top\mathbf{x}$ used by ADMM. Manual analytical gradients achieved 179 μs/iter (800 iters, 285 ms cold). Warm-start required 9.7 ms (30 iters). This variant enabled direct apples-to-apples comparison against ADMM's optimization objective and confirmed that the parametric parameterization is inherently artifact-free regardless of convergence quality.

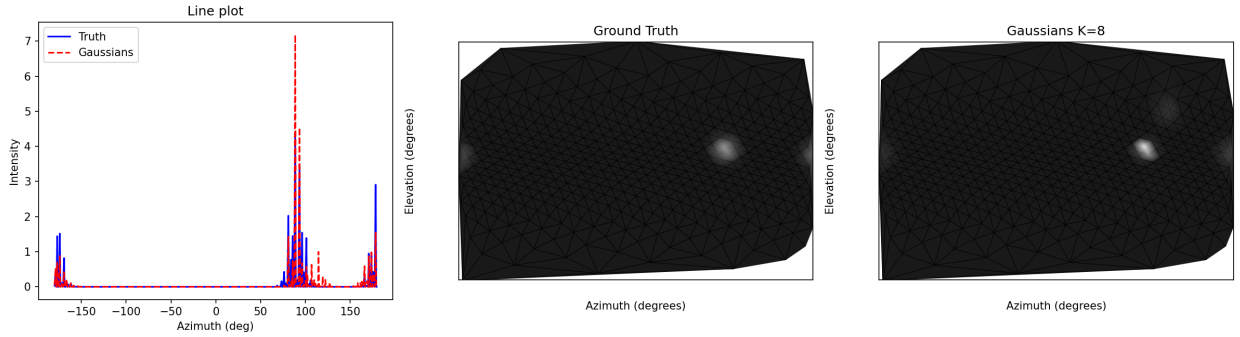


Figure 5: Grid-based variant: azimuthal intensity profiles (left) and spherical Miller-projection maps (center: ground truth, right: Gaussian reconstruction).

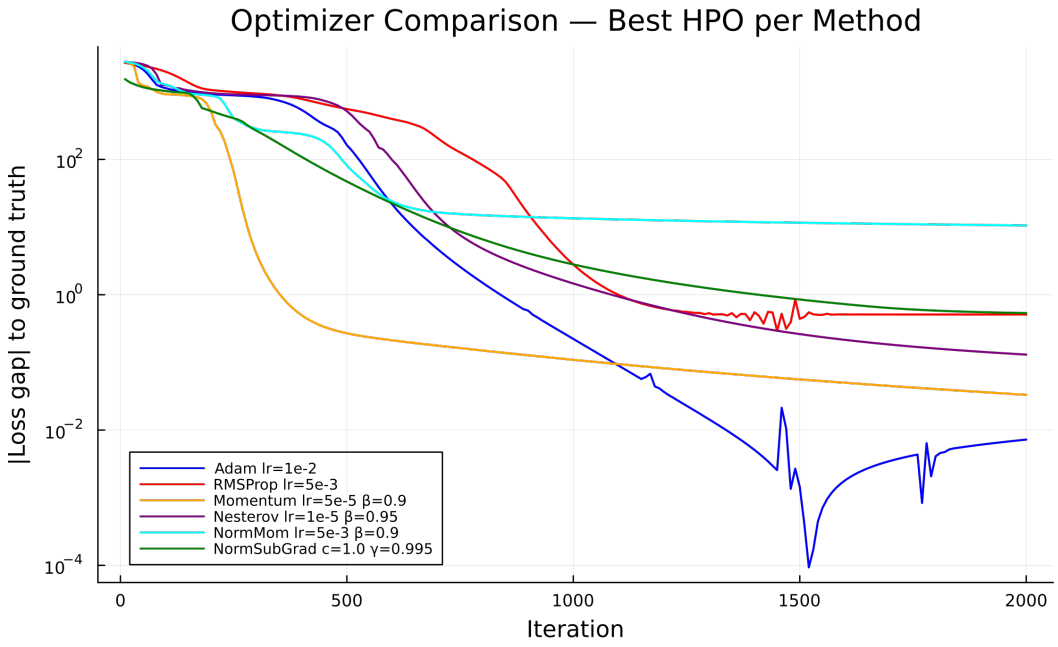


Figure 6: Optimizer comparison on the grid-based variant (K=8, 2000 iterations). Adaptive methods (Adam, RMSProp) converge reliably; momentum requires a 200× smaller LR.

Six optimization methods were benchmarked (Figure 6). The key challenge was ill-conditioning across parameter scales ($\mu_k \sim$ radians, π_k spanning orders of magnitude), resolved by Adam's per-parameter adaptive learning rates.

The direct formulation's optimizer comparison (Figure 7) confirms that the 4×4 covariance space is better conditioned than the 484-point grid.

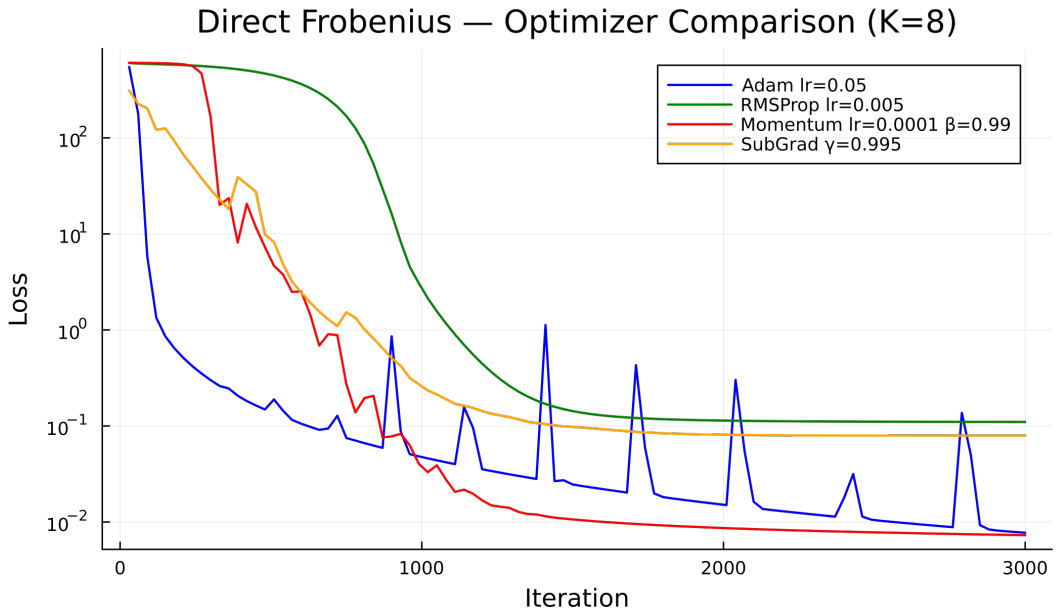


Figure 7: Optimizer comparison on the direct covariance matching formulation (K=8, 3000 iterations). The better-conditioned loss landscape allows momentum to converge with a 10× larger learning rate than in the grid variant.

B Wishart Log-Likelihood Alternative

The Frobenius loss treats all covariance entries equally. A principled alternative under the complex Gaussian measurement model is the Wishart negative log-likelihood:

$$\mathcal{L}_{\text{Wishart}}(\Theta) = \log\det C_{\text{pred}}(\Theta) + \text{tr}\left(C_{\text{pred}}^{-1}\, C_{\text{obs}}\right) \tag{8}$$

with analytical gradients via matrix derivatives. Empirically, the Wishart loss achieved NMSE 1.06 at 44 μs/iter (3× slower due to the 4×4 matrix inverse), performing worse than the Frobenius approximation (NMSE 0.40) on our synthetic test setup. The Frobenius loss is retained as the primary method for its speed-accuracy trade-off.

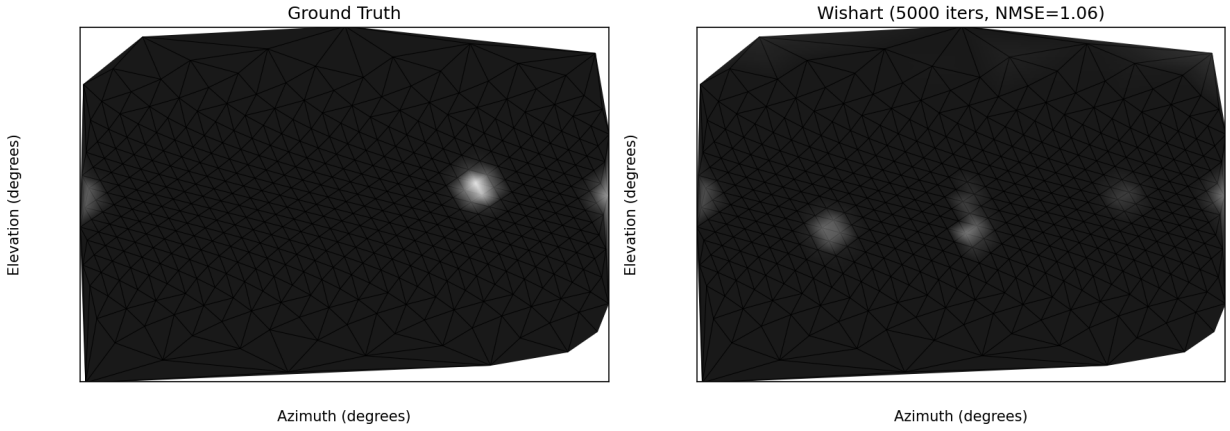


Figure 8: Wishart log-likelihood fit: ground truth (left) vs. fitted Gaussians (right). Sources partially recovered (NMSE 1.06).

C Gauss-Newton Failure Analysis

For the nonlinear least squares problem $\mathcal{L}_{\text{spatial}}$, we attempted a Gauss-Newton solver with Levenberg-Marquardt damping. The Jacobian $J \in \mathbb{R}^{32\times32}$ was validated analytically (error 0.0 against finite differences). However, the update $\Delta v = -(J^\top J + \lambda I)^{-1} J^\top r(v)$ consistently rejected all steps: the exponential nonlinearity in the steering vector $\mathbf{a}(\mu) = \exp(-ik\mathbf{r} \cdot \mathbf{s}(\mu))$ causes the linearized model to overshoot by > 2 radians per step, even with heavy damping ($\lambda \to \infty$). At large λ, the method reduces to slow gradient descent, defeating its purpose. The 4-mic array spans approximately one full wavelength (85 mm at $\lambda \approx 70$ mm), so a 2-radian step moves the source through a complete steering vector oscillation—far outside the linear regime required by Newton-type methods.

Acknowledgments

References